



SYSTEM ANALYSIS & DESIGN

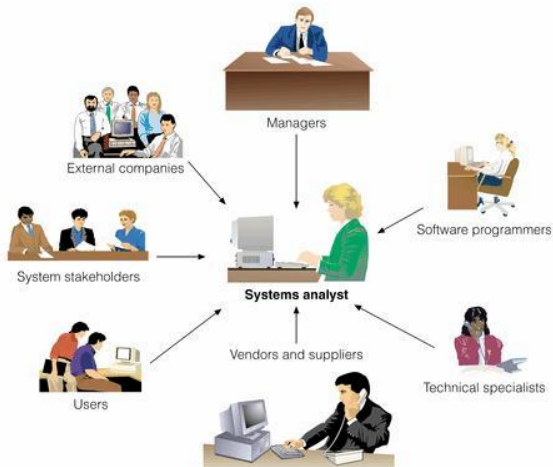
Management Information System Department
3rd Year - 2nd Semester

Lec #2

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SADAT ACADEMY FOR MANAGEMENT SCIENCES

One or More Systems Analyst!!



1 System analyst role:

- Focuses on the IS issues surrounding the system.
- Have significant training and experience in analysis and design and in programming.

2 Business analyst role:

- Focuses on the business issues surrounding the system.
- Helps to identify the business value that the system will create.

3 Requirements analyst role:

- Focuses on eliciting the requirements from the stakeholders associated with the new system.
- Highly skilled in an array of requirements elicitation techniques.

4 Infrastructure analyst role:

- Focuses on technical issues surrounding the ways the system will interact with the organization's technical infrastructure (hardware, software, networks, and databases).

5 Change management analyst role:

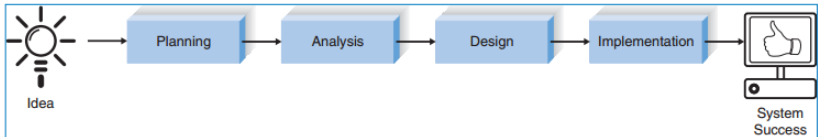
- Focuses on the people and management issues surrounding the system installation.

6 Project manager role:

- Ensures that the project is completed on time and within budget and that the system delivers the expected value to the organization.

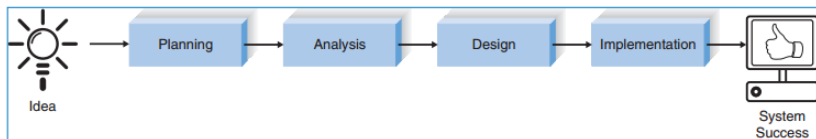
System Development Life Cycle (SDLC)

- The process of determining how an Information System (IS) can support business needs, designing the system, building it, and delivering it to users.
- Steps from the idea to validate the idea in a computer-based software is included in SDLC.



- The project:
 - Moves systematically through phases where each phase has a standard set of outputs.
 - Produces project deliverables.
 - Uses deliverables in implementation.
 - Results in actual information system.
 - Uses gradual refinement.
- Major Attributes:
 - **Planning:** Why build the system?
 - **Analysis:** Who, what, when, where will the system be?
 - **Design:** How will the system work?
 - **Implementation:** System delivery.

System Development Life Cycle



- **Methodology:** a formalized approach or series of steps.
- Each phase is itself composed of a series of steps, which rely on techniques that produce deliverables.
- Different projects may emphasize different parts of the SDLC or approach the SDLC phases in different ways, but all projects have elements of these four phases.

- The fundamental process of understanding why an IS should be built and determining how the project team will go about building it.
- Steps:
 - 1 Project Initiation:**
 - Identify business problem/opportunity.
 - Define business value.
 - Prepare system request.
 - 2 Feasibility Analysis:**
 - Technical feasibility.
 - Economic feasibility.
 - Operational feasibility.
 - Schedule feasibility.
 - 3 Project Management Planning:**
 - Work Breakdown Structure (WBS).
 - Risk identification.
 - Resource allocation.

- Deliverables:
 - Project Charter.
 - System Request.
 - Feasibility Report.
 - Risk Assessment.
 - Work Plan (Gantt Chart).
- Phase Gate: Steering Committee Approval.

- Answers the questions of who will use the system, what the system will do, and where and when it will be used.
- Steps:
 - 1 Analysis Strategy:**
 - Study the As-Is system.
 - Define the To-Be system.
 - 2 Requirements Gathering:**
 - Interviews, questionnaires, observation.
 - Functional requirements.
 - Non-functional requirements.
 - 3 Requirements Validation:**
 - Review sessions.
 - Prototyping.
 - Traceability matrix.
- Deliverables:
 - System Proposal.
 - Software Requirements Specification (SRS).
 - Use Case Diagrams.
 - Data Flow Diagrams (DFD).

- Decides how the system will operate in terms of the hardware, software, and network infrastructure that will be in place.
- Steps:
 - 1 Design Strategy:**
 - Build in-house.
 - Outsource.
 - Hybrid solution.
 - 2 Architectural Design:**
 - Layered architecture.
 - Client-server.
 - Microservices.
- Deliverables:
 - Architecture Document.
 - Database ERD.
 - UI Prototypes.
 - HLD and LLD.
 - API Specifications.
- Phase Gate: Design Review Approval.

Implementation Phase

- Build, test, and deploy the system.

- Steps:

- 1 System Construction:**

- Coding.
- Unit testing.
- Integration testing.

- 2 System Testing:**

- System testing.
- User acceptance testing (UAT).

- 3 Deployment:**

- Direct cutover.
- Parallel run.
- Phased implementation.

- Deliverables:

- Source Code.
- Test Reports.
- Deployment Guide.
- Training Materials.
- Maintenance Plan.